

GAME EDUKASI · UNITY 3D · TUTORIAL LENGKAP

EduQuest

Jelajah Sistem Organ Manusia

Panduan Lengkap Klik-demi-Klik: Membuat Elemen UI · Posisi & Skala · Warna · Kamera · Lampu · Kode Program
Third-Person · URP · Unity 6

Penulis: Putu Aurelia Lavanya (2315051035)

Mata Kuliah: Praktik Pengembangan Game Pembelajaran

Tanggal: 25 June 2026

Daftar Isi

1. Tentang Game & Tujuan
2. Persiapan Project (Unity, URP, Package)
3. Struktur Folder & Cara Membuat Script
4. Palet Warna (referensi hex)
5. **Dasar Membuat Elemen UI** (Canvas, Panel, Text, Button, Slider, Anchor, OnClick)
6. Modul Main Menu — langkah + nilai + kode
7. Modul Soal (ScriptableObject)
8. Modul Level — langkah lengkap + nilai eksak + kode
9. Modul EndScreen
10. Build Settings & WebGL
11. Troubleshooting

1. Tentang Game & Tujuan

EduQuest: Jelajah Sistem Organ Manusia — game edukasi eksplorasi + quiz untuk siswa SMP–SMA. Pemain third-person menjelajahi ruangan organ, menemukan stasiun quiz, menjawab soal.

Aspek	Keterangan
Engine	Unity 6 (URP)
Perspektif	Third-Person (Starter Assets + Cinemachine)
Input / UI	Unity Input System (baru) / TextMeshPro
Level	Jantung #8B0000 · Paru-paru #4FC3F7 · Otak #7B1FA2

Aturan: 3 stasiun/level. Benar +10, salah –1 nyawa, semua benar bonus +20, item +5. Nyawa awal 3.

MainMenu → Level1_Jantung → Level2_ParuParu → Level3_Otak → EndScreen

2. Persiapan Project

Unity 6, template **3D (URP)**. Install via Package Manager:

Package	Fungsi
Universal RP	Render pipeline
Input System	Input modern (restart bila diminta)
TextMeshPro	Teks UI — Import TMP Essentials
Cinemachine	Kamera third-person
Starter Assets - ThirdPerson	PlayerArmature + PlayerFollowCamera

Project Settings > Player > **Active Input Handling = Input System Package**. Hanya boleh ada SATU class `ThirdPersonController` .
Import TMP Essentials.

3. Struktur Folder & Cara Membuat Script

```
Assets/  
├─ Scenes/           (MainMenu, Level1-3, EndScreen)  
├─ ScriptableObjects/ (Quiz_Jantung/ParuParu/Otak)  
└─ Scripts/  
    ├─ Core/         GameManager, SceneLoader  
    ├─ Data/         QuizData  
    ├─ Gameplay/     IInteractable, PlayerInteraction, QuizStation, QuizManager,  
    │                 LevelManager, Collectible, Rotator, FloatBob, Billboard, SimpleFollowCamera  
    └─ UI/           UIManager, MainMenuController, PauseMenu, EndScreenController
```

Buat folder: klik kanan Project > Create > Folder. Buat script: klik kanan folder > Create > C# Script (nama file = nama class), buka, hapus isi bawaan, salin kode dari modul terkait.

4. Palet Warna (Referensi Hex)

Pemakaian	Jantung	Paru-paru	Otak
Dasar organ	 #8B0000	 #4FC3F7	 #7B1FA2
Dinding (×0.85)	 #760000	 #43A6D2	 #681B8A
Aksen/ceiling/pilar	 #4C0000	 #2B6B88	 #441159
Lantai	 #3E0000	 #23586F	 #370E49
Fog	 #5A0000	 #3380A1	 #501569
Glow stasiun	 #B35A5A	 #8DD8FA	 #A96DC3

UI	Hex
BG Menu / Panel Menu	#0D0D14 / #0F0F1A
Overlay panel	Hitam Alpha 0.85
Kotak quiz/feedback / result	#1F1F2E / #1A2433
Tombol jawaban / Lives / Skor / Item	#E6E6F2 / #FF4D4D / #FFE666 / #FFD933

5. Dasar Membuat Elemen UI di Unity

Bab ini menjelaskan cara membuat tiap elemen UI klik-demi-klik. Modul-modul setelah ini akan merujuk ke sini, jadi pahami dulu bagian ini.

5.1 Membuat Canvas

1. Hierarchy > klik kanan > **UI** > **Canvas**. Objek **Canvas** + **EventSystem** otomatis dibuat.
2. Pilih Canvas. Pada komponen **Canvas Scaler: UI Scale Mode** = *Scale With Screen Size*; **Reference Resolution** = X 1920, Y 1080; **Match** = 0.5.
3. Pilih **EventSystem**; jika muncul tombol kuning "Replace with InputSystemUIInputModule", klik (wajib agar tombol bisa diklik dengan Input System baru).

5.2 RectTransform & Anchor (penting)

Semua objek UI memakai **RectTransform** (bukan Transform biasa). Bagian-bagiannya:

- **Anchor Presets**: kotak bergambar di kiri-atas RectTransform. Klik untuk memilih titik jangkar (kiri-atas, tengah, dll). Tahan **Alt** saat memilih = sekaligus pindahkan objek ke sana.
- **Pos X / Pos Y**: posisi relatif terhadap anchor. **Width / Height**: ukuran kotak.
- **Stretch penuh** (mis. background): buka Anchor Presets, tahan **Alt**, pilih ikon pojok kanan-bawah (stretch-stretch). Lalu set Left/Top/Right/Bottom = 0.

Istilah anchor di tutorial: **TL**=kiri-atas, **TR**=kanan-atas, **TC**=tengah-atas, **BC**=tengah-bawah, **C**=tengah.

5.3 Membuat Panel / Image (kotak warna)

1. Klik kanan Canvas > **UI** > **Image** (atau **Panel** untuk latar semi-transparan).
2. Atur ukuran/posisi via RectTransform (5.2).
3. Warna: komponen **Image** > **Color** > klik kotak warna > isi kolom **Hexadecimal** (mis. 0F0F1A). Untuk transparan, atur nilai **A** (Alpha 0–255; $0.85 \approx 217$).

5.4 Membuat Teks (TextMeshPro)

1. Klik kanan Canvas > **UI** > **Text - TextMeshPro**. (Pertama kali muncul jendela *Import TMP Essentials* → klik Import.)
2. Pada komponen **TextMeshPro - Text (UI)**: isi kotak **Text**; atur **Font Size**; **Alignment** (kiri/tengah/kanan + atas/tengah/bawah); warna di **Vertex Color** (atau Color).

5.5 Membuat Button + Elemennya

1. Klik kanan Canvas > **UI** > **Button - TextMeshPro**.
2. Strukturnya: objek **Button** (punya komponen **Image** = latar tombol, dan komponen **Button**) dengan satu anak **Text (TMP)** = tulisan tombol.
3. **Ganti tulisan**: pilih anak **Text (TMP)** > ubah kotak Text (mis. "Main").
4. **Ganti warna tombol**: pilih objek Button > **Image** > **Color** (mis. #E6E6F2).
5. **Ukuran & posisi**: pilih objek Button > atur RectTransform.
6. **Rename** objek Button (F2) sesuai kebutuhan (mis. PlayButton).

5.6 Menyambungkan OnClick (sangat penting)

1. Pilih objek Button. Di komponen **Button**, cari kotak **On Click ()** di bawah.
2. Klik tombol + → muncul baris kosong "Runtime Only / None (Object) / No Function".
3. **Drag** objek yang berisi script tujuan (mis. **MainMenuController** dari Hierarchy) ke slot **None (Object)**.
4. Klik dropdown **No Function** → pilih nama script (mis. MainMenuController) → pilih fungsinya (mis. **OnPlay ()**). Untuk fungsi tanpa argumen, ada di bagian atas daftar.

5.7 Membuat Slider (volume)

1. Klik kanan Canvas > **UI** > **Slider**. Otomatis berisi Background, Fill Area > Fill, Handle.
2. Pada komponen **Slider**: **Min Value** 0, **Max Value** 1, **Value** 1.

5.8 Trik Cepat

Aksi	Cara
Duplikat objek	pilih objek lalu Ctrl + D
Rename	F2
Aktif/nonaktif objek	checkbox kiri-atas Inspector (sebelah nama)
Menjadikan child	drag objek ke objek lain di Hierarchy
Add Component (script)	Inspector > tombol Add Component > ketik nama

6. Modul Main Menu

A. Langkah Membangun

File > New Scene (Basic URP) > simpan **MainMenu**. Pilih Main Camera: Clear Flags = Solid Color, Background **#0D0D14**. Buat Canvas (lihat 5.1). Create Empty **GameManagers** (Add Component GameManager + SceneLoader). Create Empty **MainMenuController** (Add Component MainMenuController).

Contoh LENGKAP membuat satu tombol (PlayButton)

1. Klik kanan Canvas > UI > **Button - TextMeshPro**. Rename objeknya jadi **PlayButton** (F2).
2. Buka anak **Text (TMP)**, ubah kotak Text menjadi **Main**, Alignment Center, Font Size ~30.
3. Pilih PlayButton > RectTransform: Anchor Presets (tahan Alt) pilih **Center**; Pos X 0, Pos Y 60; Width 450, Height 80.
4. Pilih PlayButton > Image > Color = #E6E6F2.
5. Komponen Button > **On Click ()** > klik + > drag **MainMenuController** ke slot Object > dropdown > MainMenuController > **OnPlay ()**.

Ulangi cara yang sama untuk tombol lain (tinggal ganti teks, Pos Y, dan fungsi OnClick).

MainPanel (Image #0F0F1A, stretch penuh)

Elemen	Anchor	Pos	Size	Font	OnClick / Isi
Judul (Text TMP)	TC	0, -120	1200×130	90	"EduQuest" putih
Subjudul (Text TMP)	TC	0, -240	1200×60	40	#B3D9FF
PlayButton	C	0, 60	450×80	30	OnPlay
SettingsButton	C	0, -40	450×80	30	ShowSettings
AboutButton	C	0, -140	450×80	30	ShowAbout
ExitButton	C	0, -240	450×80	30	OnExit

SettingsPanel (#0F0F1A, nonaktif awal)

Elemen	Anchor	Pos	Size	Font
Judul "Pengaturan"	TC	0, -140	1000×100	70
Label "Volume"	C	0, 80	600×60	40
Slider Volume	C	0, 10	600×30	—
Info kontrol	C	0, -120	900×120	28
Button Kembali (→ShowMain)	C	0, -320	450×80	30

AboutPanel (#0F0F1A, nonaktif awal)

Elemen	Anchor	Pos	Size	Font
Judul "Tentang Game"	TC	0, -120	1000×100	70
Deskripsi	C	0, 20	1100×500	30
Button Kembali (→ShowMain)	C	0, -360	450×80	30

Assign ke MainMenuController: mainPanel, settingsPanel, aboutPanel, volumeSlider. Nonaktifkan Settings & About panel (checkbox). Simpan scene.

B. Kode Modul Main Menu

```
Assets/Scripts/UI/MainMenuController.cs
```

Logika tombol menu + volume.

```

using UnityEngine;
using UnityEngine.UI;
using EduQuest.Core;

namespace EduQuest.UI
{
    /// <summary>
    /// Mengontrol Main Menu sesuai GDD: Play, Settings, About, Exit.
    /// Juga menampilkan panel Settings (volume) dan About.
    /// </summary>
    public class MainMenuController : MonoBehaviour
    {
        [Header("Panel")]
        public GameObject mainPanel;
        public GameObject settingsPanel;
        public GameObject aboutPanel;

        [Header("Settings")]
        [Tooltip("Slider volume master (0 - 1)")]
        public Slider volumeSlider;

        [Header("Scene Tujuan")]
        public string firstLevelScene = "Level1_Jantung";

        private const string VolumeKey = "EduQuest_Volume";

        private void Start()
        {
            ShowMain();

            // Muat & terapkan volume tersimpan
            float vol = PlayerPrefs.GetFloat(VolumeKey, 1f);
            Audiolistener.volume = vol;
            if (volumeSlider != null)
            {
                volumeSlider.value = vol;
                volumeSlider.onValueChanged.AddListener(SetVolume);
            }

            // Pastikan kursor terlihat di menu
            Cursor.lockState = CursorLockMode.None;
            Cursor.visible = true;

            // Reset waktu jika datang dari pause
            Time.timeScale = 1f;
        }

        // ---- Tombol utama ----

        /// <summary>Tombol Play: reset data game lalu masuk level pertama.</summary>
        public void OnPlay()
        {
            GameManager.Instance?.ResetGame();
            if (SceneLoader.Instance != null)
                SceneLoader.Instance.LoadScene(firstLevelScene);
        }

        public void OnExit()
        {
            Debug.Log("Keluar dari game.");
        }
    }
}

```

```

        Application.Quit();
        // Catatan: Application.Quit tidak berefek di Editor maupun WebGL.
    }

    // ---- Navigasi panel ----

    public void ShowMain()
    {
        SetPanels(main: true, settings: false, about: false);
    }

    public void ShowSettings()
    {
        SetPanels(main: false, settings: true, about: false);
    }

    public void ShowAbout()
    {
        SetPanels(main: false, settings: false, about: true);
    }

    private void SetPanels(bool main, bool settings, bool about)
    {
        if (mainPanel) mainPanel.SetActive(main);
        if (settingsPanel) settingsPanel.SetActive(settings);
        if (aboutPanel) aboutPanel.SetActive(about);
    }

    // ---- Settings ----

    public void SetVolume(float value)
    {
        Audiolistener.volume = value;
        PlayerPrefs.SetFloat(VolumeKey, value);
    }
}

```

Assets/Scripts/Core/GameManager.cs

Singleton skor/nyawa/progres.

```

using System.Collections.Generic;
using UnityEngine;

namespace EduQuest.Core
{
    /// <summary>
    /// GameManager bersifat Singleton dan persist antar scene (DontDestroyOnLoad).
    /// Bertugas menyimpan skor, nyawa, dan progres level yang sudah diselesaikan.
    /// </summary>
    public class GameManager : MonoBehaviour
    {
        // Instance tunggal yang bisa diakses dari mana saja
        public static GameManager Instance { get; private set; }

        [Header("Pengaturan Awal")]
        [Tooltip("Jumlah nyawa default saat game dimulai")]
        public int startingLives = 3;

        // Data runtime
    }
}

```

```

public int Score { get; private set; }
public int Lives { get; private set; }

// Statistik jawaban (untuk result screen)
public int CorrectAnswers { get; private set; }
public int WrongAnswers { get; private set; }

// Daftar nama level yang sudah diselesaikan
private readonly HashSet<string> completedLevels = new HashSet<string>();

// Event sederhana supaya UI bisa update otomatis saat skor / nyawa berubah
public System.Action OnStatsChanged;

private void Awake()
{
    // Pola Singleton: kalau sudah ada instance lain, hancurkan duplikat
    if (Instance != null && Instance != this)
    {
        Destroy(gameObject);
        return;
    }

    Instance = this;
    DontDestroyOnLoad(gameObject);

    ResetGame();
}

/// <summary>Reset seluruh data game ke kondisi awal (dipakai saat New Game / Retry total).</summary>
public void ResetGame()
{
    Score = 0;
    Lives = startingLives;
    CorrectAnswers = 0;
    WrongAnswers = 0;
    completedLevels.Clear();
    OnStatsChanged?.Invoke();
}

/// <summary>Tambah skor sebesar amount.</summary>
public void AddScore(int amount)
{
    Score += amount;
    OnStatsChanged?.Invoke();
}

/// <summary>Catat jawaban benar (+10 skor) dan update statistik.</summary>
public void RegisterCorrectAnswer()
{
    CorrectAnswers++;
    AddScore(10);
}

/// <summary>Catat jawaban salah (kurangi 1 nyawa) dan update statistik.</summary>
public void RegisterWrongAnswer()
{
    WrongAnswers++;
    Lives = Mathf.Max(0, Lives - 1);
    OnStatsChanged?.Invoke();
}

```

```

    /// <summary>Bonus +20 jika seluruh soal dalam satu level dijawab benar.</summary>
    public void AddPerfectBonus()
    {
        AddScore(20);
    }

    /// <summary>Tandai sebuah level sudah selesai.</summary>
    public void MarkLevelCompleted(string levelName)
    {
        if (!string.IsNullOrEmpty(levelName))
            completedLevels.Add(levelName);
    }

    public bool IsLevelCompleted(string levelName) => completedLevels.Contains(levelName);

    /// <summary>Kondisi kalah: nyawa habis.</summary>
    public bool IsGameOver => Lives <= 0;

    /// <summary>Kondisi menang dipanggil saat semua level target selesai (dicek oleh SceneLoader/UI).</summary>
    public int CompletedLevelCount => completedLevels.Count;
}
}

```

Assets/Scripts/Core/SceneLoader.cs

Pindah scene + fade.

```

using System.Collections;
using UnityEngine;
using UnityEngine.SceneManagement;
using UnityEngine.UI;

namespace EduQuest.Core
{
    /// <summary>
    /// Mengatur perpindahan scene beserta transisi fade in/out sederhana.
    /// Urutan scene: MainMenu > Level1_Jantung > Level2_ParuParu > Level3_Otak > EndScreen
    /// PENTING: daftarkan semua scene tersebut di Build Settings (File > Build Settings).
    /// </summary>
    public class SceneLoader : MonoBehaviour
    {
        public static SceneLoader Instance { get; private set; }

        [Header("Transisi Fade")]
        [Tooltip("CanvasGroup atau Image hitam full-screen untuk efek fade. Boleh dikosongkan (langsung pindah tanpa fade).")]
        public CanvasGroup fadeCanvasGroup;

        [Tooltip("Durasi fade (detik)")]
        public float fadeDuration = 0.5f;

        // Urutan scene game. Pastikan nama persis sama dengan nama file scene di Build Settings.
        public static readonly string[] SceneOrder =
        {
            "MainMenu",
            "Level1_Jantung",
            "Level2_ParuParu",
            "Level3_Otak",
            "EndScreen"
        };

        private void Awake()

```

```

{
    if (Instance != null && Instance != this)
    {
        Destroy(gameObject);
        return;
    }
    Instance = this;
    DontDestroyOnLoad(gameObject);

    // Pastikan layar mulai dalam keadaan terang (alpha 0)
    if (fadeCanvasGroup != null)
    {
        fadeCanvasGroup.alpha = 0f;
        fadeCanvasGroup.blocksRaycasts = false;
    }
}

/// <summary>Muat scene berdasarkan nama, dengan transisi fade.</summary>
public void LoadScene(string sceneName)
{
    StartCoroutine(LoadSceneRoutine(sceneName));
}

/// <summary>Muat scene berikutnya sesuai urutan SceneOrder.</summary>
public void LoadNextScene()
{
    string current = SceneManager.GetActiveScene().name;
    int index = System.Array.IndexOf(SceneOrder, current);

    if (index >= 0 && index < SceneOrder.Length - 1)
    {
        LoadScene(SceneOrder[index + 1]);
    }
    else
    {
        {
            // Sudah di scene terakhir, kembali ke EndScreen sebagai fallback
            LoadScene("EndScreen");
        }
    }
}

public void LoadMainMenu() => LoadScene("MainMenu");

/// <summary>Muat ulang scene yang sedang aktif (untuk tombol Retry).</summary>
public void ReloadCurrentScene()
{
    LoadScene(SceneManager.GetActiveScene().name);
}

private IEnumerator LoadSceneRoutine(string sceneName)
{
    {
        // Fade out (layar menjadi gelap)
        yield return Fade(1f);

        // Muat scene secara asinkron
        AsyncOperation op = SceneManager.LoadSceneAsync(sceneName);
        while (op != null && !op.isDone)
            yield return null;

        // Fade in (layar terang kembali)
        yield return Fade(0f);
    }
}

```

```
private IEnumerator Fade(float targetAlpha)
{
    if (fadeCanvasGroup == null)
        yield break;

    fadeCanvasGroup.blocksRaycasts = true;
    float startAlpha = fadeCanvasGroup.alpha;
    float time = 0f;

    while (time < fadeDuration)
    {
        time += Time.unscaledDeltaTime; // unscaled supaya tetap jalan walau Time.timeScale = 0
        fadeCanvasGroup.alpha = Mathf.Lerp(startAlpha, targetAlpha, time / fadeDuration);
        yield return null;
    }

    fadeCanvasGroup.alpha = targetAlpha;
    fadeCanvasGroup.blocksRaycasts = targetAlpha > 0.9f;
}
}
```

7. Modul Soal (ScriptableObject)

A. Langkah Membuat Aset

- 1. Folder Assets/ScriptableObjects > klik kanan > **Create > EduQuest > Quiz Data**. Beri nama Quiz_Jantung. Ulangi untuk Quiz_ParuParu & Quiz_Otak.
- 2. Pilih aset > Inspector: isi **Organ Name**; pada **Questions** klik + untuk menambah 3 elemen; tiap elemen isi Question Text, 4 Options, Correct Index (0–3), Explanation.

Aset	Soal & jawaban benar (index)
Quiz_Jantung	(1) Jumlah ruang? → 4 (2). (2) Pembuluh keluar? → Arteri (1). (3) Pemompa ke tubuh? → Bilik kiri (2).
Quiz_ParuParu	(1) Pertukaran gas? → Alveolus (2). (2) Saluran utama? → Trakea (1). (3) Otot napas? → Diafragma (2).
Quiz_Otak	(1) Keseimbangan? → Otak kecil (1). (2) Berpikir? → Otak besar (2). (3) Detak jantung? → Batang otak (0).

B. Kode Modul Soal

Assets/Scripts/Data/QuizData.cs

ScriptableObject soal.

```
using System.Collections.Generic;
using UnityEngine;

namespace EduQuest.Data
{
    /// <summary>
    /// Struktur data untuk satu butir soal pilihan ganda.
    /// Ditandai [System.Serializable] supaya bisa muncul & diisi lewat Inspector.
    /// </summary>
    [System.Serializable]
    public class Question
    {
        [Tooltip("Teks pertanyaan yang ditampilkan ke pemain")]
        [TextArea(2, 4)]
        public string questionText;

        [Tooltip("4 pilihan jawaban (index 0 - 3)")]
        public string[] options = new string[4];

        [Tooltip("Index jawaban yang benar (0, 1, 2, atau 3)")]
        [Range(0, 3)]
        public int correctIndex;

        [Tooltip("Penjelasan singkat yang muncul setelah pemain menjawab")]
        [TextArea(2, 4)]
        public string explanation;
    }

    /// <summary>
    /// ScriptableObject berisi kumpulan soal untuk satu organ / satu level.
    /// Buat asset melalui menu: Create > EduQuest > Quiz Data
    /// </summary>
    [CreateAssetMenu(fileName = "Quiz_Organ", menuName = "EduQuest/Quiz Data", order = 0)]
    public class QuizData : ScriptableObject
    {
        [Tooltip("Nama organ / nama level, contoh: Jantung")]
        public string organName = "Organ";
    }
}
```



```
[Tooltip("Daftar soal untuk organ ini")]  
public List<Question> questions = new List<Question>();  
}  
}
```

8. Modul Level (Langkah Lengkap + Nilai Eksak)

Contoh **Level1_Jantung** (merah). Level lain: ganti warna/quiz/fog. New Scene > simpan **Level1_Jantung**.

8.1 Lingkungan, Fog & Skybox

Directional Light	Nilai
Rotation	50, -30, 0
Intensity / Color / Shadow	0.5 / #DCB3B3 / Soft

Window > Rendering > Lighting > Environment: Fog **Exponential Squared**, Color #5A0000 (Paru #3380A1, Otak #501569), Density 0.045 (Paru 0.028, Otak 0.035). Ambient (Color/Flat) #530000. Skybox = Material shader Skybox/Procedural, Sky Tint warna organ, Exposure 0.7.

8.2 Ruang (parent Empty "Room_Jantung")

Buat objek: GameObject > 3D Object > Cube. Buat Material (klik kanan Project > Create > Material), set Base Map / Color via hex, lalu drag material ke objek.

Cube	Position	Scale	Material
Floor	0, -0.2, 0	20, 0.4, 36	#3E0000
Ceiling	0, 6.2, 0	20, 0.4, 36	#4C0000
Wall_North	0, 3, 18	20, 6, 0.4	#760000
Wall_South	0, 3, -18	20, 6, 0.4	#760000
Wall_East	10, 3, 0	0.4, 6, 36	#760000
Wall_West	-10, 3, 0	0.4, 6, 36	#760000

4 Point Light (Range 16, Intensity 2.2, #CB8C8C) Y=5.2, Z = -10.8 / -3.6 / 3.6 / 10.8. **8 Pilar** (Cube 0.8×6×0.8, #4C0000) X=±8.8, Z = -13 / -5 / 3 / 11.

Objek	Position	Scale
PlayerSpawnPoint (Empty)	0, 0.1, -15.5	—
OrganPlatform (Cube #4C0000)	0, 0.25, 14	6, 0.5, 6
OrganModel_Holder (Empty)	0, 1.8, 14	—

8.3 Model Organ (di Holder, posisi LOKAL; + Rotator Y25)

Jantung: 2 Sphere (±0.45,0.4,0 s1) + Sphere badan (0,-0.3,0 s1.3,1.6,1.3) + 2 Cylinder arteri (±0.3,1.1,0 s0.25,0.5,0.25). **Paru:** 2 Sphere lobus (±0.7,0,0 s1.1,1.9,1) + Cylinder trakea (0,1.1,0 s0.3,0.6,0.3) + 2 bronkus (±0.35,0.5,0 s0.18,0.4,0.18). **Otak:** Sphere utama (0,0.2,0 s1.8,1.4,1.6) + 3 gyrus (s0.6) + cerebellum (0,-0.5,-0.7 s0.7).

8.4 Player & Kamera

Langkah	Detail
Player	Drag PlayerArmature ke 0,0.1,-15.5; Tag = Player ; Add PlayerInteraction (isi ThirdPersonController, StarterAssetsInputs, PlayerInput)
Kamera	Main Camera Add CinemachineBrain ; drag PlayerFollowCamera , Follow = PlayerCameraRoot

8.5 Manager + Stasiun + Collectible + AI Guide

GameManagers (GameManager + SceneLoader + Canvas hitam ber-CanvasGroup → fadeCanvasGroup). **QuizManager** (Player Interaction = PlayerArmature). **LevelManager** (Quiz Data = Quiz_Jantung, Total Stations = 3).

3 QuizStation di (-5,0,-6),(5,0,1),(-5,0,8). Tiap stasiun (lokal): Pedestal Cube (0,0.5,0 s1.6,1,1.6); Orb Sphere (0,2,0 s0.9 #B35A5A) + FloatBob + Rotator Y40; Point Light (0,2,0 Range7 Int2.5); Label 3D Text (0,3.1,0 s0.1) "Soal n" + Billboard; Box Collider **Is Trigger** Center (0,1.5,0) Size

4×3×4. Add QuizStation (Quiz Data, Question Index 0/1/2, Station Number 1/2/3, Total 3, Glow Renderer=Orb, Active Indicator=Light).

6 Collectible Sphere (s0.5 #FFD933) + Sphere Collider Is Trigger + Collectible + Rotator Y90, di (2,1,-9)(-3,1,-2)(4,1,5)(-2,1,11)(6,1,-4)(-6,1,3).

AI Guide di (2.5,1.7,-13.5): Sphere s0.6 + Rotator Y30, Point Light Range5 Int1.6, Label 3D Text + Billboard, FloatBob pada parent.

8.6 Canvas UI (GameCanvas 1920×1080; Add UIManager + PauseMenu)

Buat tiap elemen sesuai cara di Bab 5, lalu assign ke field UIManager.

HUD	Anchor	Pos	Size	Font/Warna
ScoreText	TL	40, -30	400×60	36 putih
LivesText	TR	-40, -30	400×60	40 #FF4D4D
LevelNameText	TC	0, -30	700×60	42 putih
StationProgressText	TC	0, -95	500×50	30 #CCE6FF
InteractPrompt	BC	0, 140	800×60	34 kuning

Quiz (overlay α0.85; Box #1F1F2E 1100×720)	Pos	Size	Font
Progress / QuestionText	0,-30 / 0,-180	900×50 / 980×220	30 / 40
Answer 0/1/2/3	0, -360 / -455 / -550 / -645	880×80	30

Feedback (#1F1F2E 1000×520)	Pos	Size	Font
Title / Explanation / Lanjut	0,-60 / 0,0 / 0,50	900×90 / 900×240 / 400×80	60 / 32 / —

Result (#1A2433 1000×760)	Pos	Size	Font/Warna
Title	0, -50	900×90	60
Skor / Benar / Salah	0,-170 / -250 / -310	900×70 / 60 / 60	44 #FFE666 / 36 #66FF66 / 36 #FF8080
Motivasi	0, -400	920×120	30
Next / Ulangi / Menu	0, 230 / 145 / 60	500×75	—

Pause (Canvas Sort 20; overlay α0.85)	Anchor	Pos	Size	Font
Title "JEDA"	TC	0, -180	800×120	80
Lanjutkan / Ulangi / Menu	C	0, 60 / -40 / -140	450×80	—

Assign semua elemen ke field UIManager (scoreText, livesText, levelNameText, stationProgressText, interactPrompt, quizPanel, questionText, questionProgressText, answerButtons[0..3], answerLabels[0..3], feedbackPanel, ..., resultPanel, ...). Sembunyikan quiz/feedback/result/pause panel & interactPrompt di awal.

8.7 Kode Modul Level

Assets/Scripts/Gameplay/IInteractable.cs

```
namespace EduQuest.Gameplay
{
    /// <summary>
    /// Antarmuka untuk objek yang bisa diinteraksi pemain dengan tombol E
    /// (mis. stasiun quiz). Dipakai oleh PlayerInteraction.
    /// </summary>
    public interface IInteractable
    {
        /// <summary>Apakah objek masih bisa diinteraksi saat ini.</summary>
```

```

    bool CanInteract { get; }

    /// <summary>Aksi yang dijalankan saat pemain menekan E.</summary>
    void Interact();
}
}

```

Assets/Scripts/Gameplay/PlayerInteraction.cs

Deteksi E + on/off gerakan.

```

using UnityEngine;
using UnityEngine.InputSystem; // Input System baru
using EduQuest.UI;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Mendeteksi input tombol "E" untuk berinteraksi dengan objek terdekat
    /// (mis. QuizStation). Juga mengatur enable/disable gerakan pemain
    /// (ThirdPersonController) saat quiz aktif.
    /// </summary>
    public class PlayerInteraction : MonoBehaviour
    {
        [Header("Referensi Kontroler Pemain")]
        [Tooltip("Drag komponen ThirdPersonController di sini (sebagai MonoBehaviour).")]
        public MonoBehaviour thirdPersonController;

        [Tooltip("Drag komponen StarterAssetsInputs di sini (opsional).")]
        public MonoBehaviour starterAssetsInputs;

        [Tooltip("Drag komponen PlayerInput di sini (opsional).")]
        public PlayerInput playerInput;

        // Objek yang sedang berada dalam jangkauan (null jika tidak ada)
        private IInteractable currentInteractable;

        private void Update()
        {
            // Jika quiz sedang berlangsung, abaikan input interaksi
            if (QuizManager.Instance != null && QuizManager.Instance.IsQuizActive)
                return;

            bool pressedE = Keyboard.current != null && Keyboard.current.eKey.wasPressedThisFrame;

            if (pressedE && currentInteractable != null && currentInteractable.CanInteract)
            {
                currentInteractable.Interact();
            }
        }

        /// <summary>Set objek interaksi aktif (dipanggil dari QuizStation/TriggerZone).</summary>
        public void SetInteractable(IInteractable interactable)
        {
            currentInteractable = interactable;
        }

        /// <summary>Hapus objek interaksi jika cocok.</summary>
        public void ClearInteractable(IInteractable interactable)
        {
            if (ReferenceEquals(currentInteractable, interactable))

```

```

        currentInteractable = null;
    }

    /// <summary>Enable / disable gerakan & kamera pemain (dimatikan saat quiz aktif).</summary>
    public void SetMovementEnabled(bool enabled)
    {
        if (thirdPersonController != null) thirdPersonController.enabled = enabled;
        if (starterAssetsInputs != null) starterAssetsInputs.enabled = enabled;
        if (playerInput != null) playerInput.enabled = enabled;
    }
}

```

Assets/Scripts/Gameplay/QuizManager.cs

Tampilkan & periksa satu soal.

```

using System;
using UnityEngine;
using EduQuest.Core;
using EduQuest.Data;
using EduQuest.UI;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Menangani tampilan & pemeriksaan SATU soal quiz (mode stasiun eksplorasi).
    /// Saat quiz aktif, gerakan pemain dimatikan. Setelah selesai, hasil benar/salah
    /// dilaporkan balik ke pemanggil (QuizStation) lewat callback.
    /// </summary>
    public class QuizManager : MonoBehaviour
    {
        public static QuizManager Instance { get; private set; }

        [Header("Referensi")]
        [Tooltip("Untuk menonaktifkan gerakan pemain saat quiz aktif")]
        public PlayerInteraction playerInteraction;

        public bool IsQuizActive { get; private set; }

        private Question currentQuestion;
        private Action<bool> onComplete;
        private bool answered;

        private void Awake()
        {
            if (Instance != null && Instance != this) Destroy(gameObject);
            else Instance = this;
        }

        /// <summary>
        /// Tampilkan satu soal. stationIndex & total dipakai untuk teks "Stasiun x / y".
        /// onComplete(true/false) dipanggil setelah pemain menutup feedback.
        /// </summary>
        public void ShowSingleQuestion(Question q, int stationIndex, int total, Action<bool> onComplete)
        {
            if (IsQuizActive || q == null) return;

            IsQuizActive = true;
            currentQuestion = q;
            this.onComplete = onComplete;

```

```

        answered = false;

        // Matikan gerakan pemain + tampilkan kursor untuk klik UI
        if (playerInteraction != null) playerInteraction.SetMovementEnabled(false);
        SetCursor(true);

        UIManager.Instance?.ShowInteractPrompt(false);
        UIManager.Instance?.ShowQuestion(q, stationIndex, total, OnAnswerSelected);
    }

    private void OnAnswerSelected(int selectedIndex)
    {
        if (answered) return;
        answered = true;

        bool isCorrect = selectedIndex == currentQuestion.correctIndex;

        // Skor / nyawa
        if (isCorrect) GameManager.Instance?.RegisterCorrectAnswer(); // +10
        else GameManager.Instance?.RegisterWrongAnswer(); // -1 nyawa

        UIManager.Instance?.HighlightAnswer(selectedIndex, currentQuestion.correctIndex);
        UIManager.Instance?.ShowFeedback(isCorrect, currentQuestion.explanation, () => Finish(isCorrect));
    }

    private void Finish(bool isCorrect)
    {
        UIManager.Instance?.HideQuiz();

        IsQuizActive = false;
        if (playerInteraction != null) playerInteraction.SetMovementEnabled(true);
        SetCursor(false);

        var cb = onComplete;
        onComplete = null;
        cb?.Invoke(isCorrect);
    }

    private void SetCursor(bool unlock)
    {
        Cursor.lockState = unlock ? CursorLockMode.None : CursorLockMode.Locked;
        Cursor.visible = unlock;
    }
}

```

Assets/Scripts/Gameplay/QuizStation.cs

Stasiun 1 soal.

```

using UnityEngine;
using EduQuest.Data;
using EduQuest.UI;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Satu stasiun quiz yang tersebar di ruangan. Pemain mendekat lalu menekan E
    /// untuk menjawab SATU soal. Setelah dijawab, stasiun dinonaktifkan dan
    /// melapor ke LevelManager.
    /// Collider pada objek ini WAJIB Is Trigger.
    
```

```

/// </summary>
[RequireComponent(typeof(Collider))]
public class QuizStation : MonoBehaviour, IInteractable
{
    [Header("Data")]
    public QuizData quizData;

    [Tooltip("Index soal yang dipakai stasiun ini")]
    public int questionIndex;

    [Tooltip("Nomor stasiun untuk ditampilkan (1-based)")]
    public int stationNumber = 1;

    [Tooltip("Total stasiun di level")]
    public int totalStations = 3;

    [Header("Visual")]
    [Tooltip("Renderer yang warnanya diubah saat selesai (mis. bola organ di stasiun)")]
    public Renderer glowRenderer;
    public Color completedColor = new Color(0.2f, 1f, 0.3f);
    [Tooltip("Objek yang dimatikan saat stasiun selesai (mis. cahaya/penanda)")]
    public GameObject activeIndicator;

    [Header("Pengaturan")]
    public string playerTag = "Player";

    private bool answered;

    public bool CanInteract => !answered && quizData != null;

    private void OnTriggerEnter(Collider other)
    {
        if (answered || !other.CompareTag(playerTag)) return;

        var pi = other.GetComponentInParent<PlayerInteraction>();
        if (pi != null) pi.SetInteractable(this);

        UIManager.Instance?.ShowInteractPrompt(true);
    }

    private void OnTriggerExit(Collider other)
    {
        if (!other.CompareTag(playerTag)) return;

        var pi = other.GetComponentInParent<PlayerInteraction>();
        if (pi != null) pi.ClearInteractable(this);

        UIManager.Instance?.ShowInteractPrompt(false);
    }

    /// <summary>Dipanggil PlayerInteraction saat pemain menekan E di dekat stasiun.</summary>
    public void Interact()
    {
        if (!CanInteract) return;

        UIManager.Instance?.ShowInteractPrompt(false);

        int idx = Mathf.Clamp(questionIndex, 0, quizData.questions.Count - 1);
        Question q = quizData.questions[idx];

        QuizManager.Instance?.ShowSingleQuestion(q, stationNumber - 1, totalStations, OnAnswered);
    }
}

```

```

    }

    private void OnAnswered(bool isCorrect)
    {
        answered = true;

        // Ubah visual menjadi "selesai"
        if (glowRenderer != null)
        {
            var mat = glowRenderer.material; // instance, aman diubah saat runtime
            mat.color = completedColor;
            if (mat.HasProperty("_BaseColor")) mat.SetColor("_BaseColor", completedColor);
            if (mat.HasProperty("_EmissionColor")) mat.SetColor("_EmissionColor", completedColor);
        }
        if (activeIndicator != null) activeIndicator.SetActive(false);

        LevelManager.Instance?.HandleStationComplete(isCorrect);
    }
}
}

```

Assets/Scripts/Gameplay/LevelManager.cs

Koordinator level + result.

```

using UnityEngine;
using EduQuest.Core;
using EduQuest.Data;
using EduQuest.UI;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Mengoordinasi satu level berbasis eksplorasi:
    /// melacak berapa stasiun quiz yang sudah diselesaikan, menghitung bonus
    /// "semua benar", dan menampilkan result screen saat semua stasiun selesai.
    /// </summary>
    public class LevelManager : MonoBehaviour
    {
        public static LevelManager Instance { get; private set; }

        [Header("Data Level")]
        public QuizData quizData;

        [Tooltip("Jumlah stasiun quiz di level ini")]
        public int totalStations = 3;

        private int answeredStations;
        private int correctStations;
        private bool levelEnded;

        private void Awake()
        {
            if (Instance != null && Instance != this) Destroy(gameObject);
            else Instance = this;
        }

        private void Start()
        {
            UIManager.Instance?.UpdateStationProgress(0, totalStations);
        }
    }
}

```



```

    /// <summary>Dipanggil oleh QuizStation setelah pemain menjawab soal stasiun itu.</summary>
    public void HandleStationComplete(bool isCorrect)
    {
        if (levelEnded) return;

        answeredStations++;
        if (isCorrect) correctStations++;

        UIManager.Instance?.UpdateStationProgress(answeredStations, totalStations);

        // Kalah jika nyawa habis
        if (GameManager.Instance != null && GameManager.Instance.IsGameOver)
        {
            EndLevel(false);
            return;
        }

        // Semua stasiun selesai -> level beres
        if (answeredStations >= totalStations)
        {
            if (correctStations >= totalStations)
                GameManager.Instance?.AddPerfectBonus(); // bonus +20
            EndLevel(true);
        }
    }

    private void EndLevel(bool passed)
    {
        levelEnded = true;

        var gm = GameManager.Instance;
        bool reallyPassed = passed && (gm == null || !gm.IsGameOver);

        if (reallyPassed && gm != null && quizData != null)
            gm.MarkLevelCompleted(quizData.orgaName);

        // Tampilkan kursor untuk klik result
        Cursor.lockState = CursorLockMode.None;
        Cursor.visible = true;

        UIManager.Instance?.ShowResult(
            reallyPassed,
            onNextLevel: () => SceneLoader.Instance?.LoadNextScene(),
            onRetry: () => SceneLoader.Instance?.ReloadCurrentScene(),
            onMainMenu: () => SceneLoader.Instance?.LoadMainMenu());
    }
}

```

Assets/Scripts/Gameplay/Collectible.cs

Item poin.

```

using UnityEngine;
using EduQuest.Core;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Item yang bisa dikumpulkan pemain sambil menjelajah ("poin pengetahuan").

```

```

/// Menambah skor lalu menghilang. Colliders harus Is Trigger.
/// </summary>
[RequireComponent(typeof(Collider))]
public class Collectible : MonoBehaviour
{
    [Tooltip("Skor yang didapat saat dikumpulkan")]
    public int scoreValue = 5;

    [Tooltip("Tag pemain")]
    public string playerTag = "Player";

    [Tooltip("Efek partikel opsional saat dikumpulkan")]
    public GameObject collectEffect;

    private bool collected;

    private void Reset()
    {
        var col = GetComponent<Collider>();
        if (col) col.isTrigger = true;
    }

    private void OnTriggerEnter(Collider other)
    {
        if (collected || !other.CompareTag(playerTag)) return;
        collected = true;

        GameManager.Instance?.AddScore(scoreValue);

        if (collectEffect != null)
            Instantiate(collectEffect, transform.position, Quaternion.identity);

        Destroy(gameObject);
    }
}

```

Assets/Scripts/Gameplay/Rotator.cs

Memutar objek.

```

using UnityEngine;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Memutar objek secara perlahan. Dipakai untuk model organ 3D agar menarik perhatian.
    /// </summary>
    public class Rotator : MonoBehaviour
    {
        [Tooltip("Kecepatan rotasi (derajat per detik) untuk tiap sumbu")]
        public Vector3 rotationSpeed = new Vector3(0f, 30f, 0f);

        private void Update()
        {
            transform.Rotate(rotationSpeed * Time.deltaTime, Space.World);
        }
    }
}

```

Assets/Scripts/Gameplay/FloatBob.cs

Efek mengambang.

```
using UnityEngine;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Membuat objek mengambang naik-turun perlahan (efek melayang).
    /// </summary>
    public class FloatBob : MonoBehaviour
    {
        [Tooltip("Jarak naik-turun")]
        public float amplitude = 0.25f;

        [Tooltip("Kecepatan ayunan")]
        public float frequency = 1f;

        private Vector3 startLocalPos;

        private void Start()
        {
            startLocalPos = transform.localPosition;
        }

        private void Update()
        {
            float y = Mathf.Sin(Time.time * frequency) * amplitude;
            transform.localPosition = startLocalPos + Vector3.up * y;
        }
    }
}
```

Assets/Scripts/Gameplay/Billboard.cs

Label menghadap kamera.

```
using UnityEngine;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Membuat objek (mis. label teks 3D) selalu menghadap kamera.
    /// </summary>
    public class Billboard : MonoBehaviour
    {
        private Camera cam;

        private void LateUpdate()
        {
            if (cam == null) cam = Camera.main;
            if (cam == null) return;

            // Arahkan +Z objek menjauh dari kamera agar teks terbaca normal
            transform.rotation = Quaternion.LookRotation(transform.position - cam.transform.position);
        }
    }
}
```

Assets/Scripts/Gameplay/SimpleFollowCamera.cs

Kamera fallback.

```
using UnityEngine;

namespace EduQuest.Gameplay
{
    /// <summary>
    /// Kamera pengikut sederhana sebagai fallback bila belum memakai
    /// PlayerFollowCamera (Cinemachine) dari Starter Assets.
    /// Mengikuti target dengan offset tetap dan sedikit smoothing.
    /// </summary>
    public class SimpleFollowCamera : MonoBehaviour
    {
        [Tooltip("Target yang diikuti (biasanya pemain)")]
        public Transform target;

        [Tooltip("Posisi kamera relatif terhadap target")]
        public Vector3 offset = new Vector3(0f, 3f, -6f);

        [Tooltip("Seberapa cepat kamera mengikuti")]
        public float smoothSpeed = 8f;

        [Tooltip("Titik yang dilihat kamera di atas target")]
        public float lookHeight = 1.5f;

        private void LateUpdate()
        {
            if (target == null) return;

            Vector3 desired = target.position + offset;
            transform.position = Vector3.Lerp(transform.position, desired, smoothSpeed * Time.deltaTime);
            transform.LookAt(target.position + Vector3.up * lookHeight);
        }
    }
}
```

Assets/Scripts/UI/UIManager.cs

HUD + panel quiz/feedback/result.

```
using System;
using UnityEngine;
using UnityEngine.UI;
using TMPro;
using EduQuest.Core;
using EduQuest.Data;

namespace EduQuest.UI
{
    /// <summary>
    /// Mengelola seluruh tampilan UI: HUD, panel quiz, feedback, dan result screen.
    /// Bersifat Singleton per-scene (tidak persist) supaya QuizManager mudah memanggilnya.
    /// </summary>
    public class UIManager : MonoBehaviour
    {
        public static UIManager Instance { get; private set; }

        [Header("HUD")]
        [Tooltip("Teks skor (kiri atas)")]
    }
```

```

public TMP_Text scoreText;
[Tooltip("Teks / kumpulan ikon hati (kanan atas)")]
public TMP_Text livesText;
[Tooltip("Nama level (tengah atas)")]
public TMP_Text levelNameText;
[Tooltip("Progres stasiun, contoh: Stasiun 1 / 3 (di bawah nama level)")]
public TMP_Text stationProgressText;
[Tooltip("Prompt interaksi 'Tekan E' ")]
public GameObject interactPrompt;

[Header("Quiz Panel")]
public GameObject quizPanel;
public TMP_Text questionText;
[Tooltip("4 tombol jawaban, urutan index 0 - 3")]
public Button[] answerButtons = new Button[4];
[Tooltip("Label teks pada masing-masing tombol jawaban")]
public TMP_Text[] answerLabels = new TMP_Text[4];
[Tooltip("Teks nomor soal, contoh: Soal 1 / 3")]
public TMP_Text questionProgressText;

[Header("Feedback Panel")]
public GameObject feedbackPanel;
public TMP_Text feedbackTitleText; // BENAR / SALAH
public TMP_Text feedbackExplanationText; // penjelasan singkat
public Button feedbackNextButton; // tombol Lanjut

[Header("Result Screen")]
public GameObject resultPanel;
public TMP_Text resultScoreText;
public TMP_Text resultCorrectText;
public TMP_Text resultWrongText;
public TMP_Text motivationText; // pesan motivasi
public Button nextLevelButton;
public Button retryButton;
public Button mainMenuButton;

[Header("Warna Feedback")]
public Color correctColor = new Color(0.2f, 0.8f, 0.2f);
public Color wrongColor = new Color(0.85f, 0.2f, 0.2f);
public Color defaultButtonColor = Color.white;

[Header("Info Level")]
[Tooltip("Nama level untuk ditampilkan di HUD, contoh: Level 1 - Jantung")]
public string levelDisplayName = "Level";

private void Awake()
{
    if (Instance != null && Instance != this)
        Destroy(gameObject);
    else
        Instance = this;
}

private void Start()
{
    // Sembunyikan semua panel di awal
    if (quizPanel) quizPanel.SetActive(false);
    if (feedbackPanel) feedbackPanel.SetActive(false);
    if (resultPanel) resultPanel.SetActive(false);
    if (interactPrompt) interactPrompt.SetActive(false);
}

```

```

        if (levelNameText) levelNameText.text = levelDisplayName;

        // Subscribe ke event GameManager supaya HUD update otomatis
        if (GameManager.Instance != null)
        {
            GameManager.Instance.OnStatsChanged += UpdateHUD;
            UpdateHUD();
        }
    }

    private void OnDestroy()
    {
        if (GameManager.Instance != null)
            GameManager.Instance.OnStatsChanged -= UpdateHUD;
    }

    // ----- HUD -----

    /// <summary>Perbarui tampilan skor dan nyawa di HUD.</summary>
    public void UpdateHUD()
    {
        var gm = GameManager.Instance;
        if (gm == null) return;

        if (scoreText) scoreText.text = $"Skor: {gm.Score}";

        if (livesText)
        {
            // Tampilkan nyawa sebagai ikon hati
            string hearts = "";
            for (int i = 0; i < gm.Lives; i++) hearts += "♥ "; // ♥
            livesText.text = hearts.Trim();
        }
    }

    /// <summary>Tampilkan / sembunyikan prompt "Tekan E untuk interaksi".</summary>
    public void ShowInteractPrompt(bool show)
    {
        if (interactPrompt) interactPrompt.SetActive(show);
    }

    /// <summary>Perbarui teks progres stasiun (berapa stasiun selesai dari total).</summary>
    public void UpdateStationProgress(int done, int total)
    {
        if (stationProgressText)
            stationProgressText.text = $"Stasiun: {done} / {total}";
    }

    // ----- Quiz Panel -----

    /// <summary>
    /// Tampilkan satu soal. Callback onAnswerSelected dipanggil dengan index jawaban yang dipilih.
    /// </summary>
    public void ShowQuestion(Question q, int currentIndex, int total, Action<int> onAnswerSelected)
    {
        if (feedbackPanel) feedbackPanel.SetActive(false);
        if (quizPanel) quizPanel.SetActive(true);

        if (questionText) questionText.text = q.questionText;
        if (questionProgressText) questionProgressText.text = $"Soal {currentIndex + 1} / {total}";
    }

```

```

        for (int i = 0; i < answerButtons.Length; i++)
        {
            int captured = i; // hindari closure trap
            var btn = answerButtons[i];
            if (btn == null) continue;

            btn.interactable = true;
            SetButtonColor(btn, defaultButtonColor);

            if (i < answerLabels.Length && answerLabels[i] != null && q.options != null && i < q.options.Length)
                answerLabels[i].text = q.options[i];

            btn.onClick.RemoveAllListeners();
            btn.onClick.AddListener(() => onAnswerSelected?.Invoke(captured));
        }
    }

    /// <summary>Berikan warna tombol: hijau untuk benar, merah untuk salah. Lalu nonaktifkan semua tombol.</summary>
    public void HighlightAnswer(int selectedIndex, int correctIndex)
    {
        for (int i = 0; i < answerButtons.Length; i++)
        {
            if (answerButtons[i] == null) continue;
            answerButtons[i].interactable = false;

            if (i == correctIndex)
                SetButtonColor(answerButtons[i], correctColor);
            else if (i == selectedIndex)
                SetButtonColor(answerButtons[i], wrongColor);
        }
    }

    private void SetButtonColor(Button btn, Color c)
    {
        var img = btn.GetComponent<Image>();
        if (img != null) img.color = c;
    }

    // ----- Feedback Panel -----

    /// <summary>Tampilkan panel feedback benar/salah beserta penjelasan.</summary>
    public void ShowFeedback(bool isCorrect, string explanation, Action onNext)
    {
        if (feedbackPanel) feedbackPanel.SetActive(true);

        if (feedbackTitleText)
        {
            feedbackTitleText.text = isCorrect ? "BENAR! ✓" : "SALAH ✗";
            feedbackTitleText.color = isCorrect ? correctColor : wrongColor;
        }

        if (feedbackExplanationText) feedbackExplanationText.text = explanation;

        if (feedbackNextButton)
        {
            feedbackNextButton.onClick.RemoveAllListeners();
            feedbackNextButton.onClick.AddListener(() => onNext?.Invoke());
        }
    }

    public void HideQuiz()

```

```

    {
        if (quizPanel) quizPanel.SetActive(false);
        if (feedbackPanel) feedbackPanel.SetActive(false);
    }

    // ----- Result Screen -----

    /// <summary>
    /// Tampilkan layar hasil di akhir level.
    /// </summary>
    public void ShowResult(bool levelPassed, Action onNextLevel, Action onRetry, Action onMainMenu)
    {
        var gm = GameManager.Instance;
        HideQuiz();
        if (resultPanel) resultPanel.SetActive(true);

        if (gm != null)
        {
            if (resultScoreText) resultScoreText.text = $"Skor Total: {gm.Score}";
            if (resultCorrectText) resultCorrectText.text = $"Benar: {gm.CorrectAnswers}";
            if (resultWrongText) resultWrongText.text = $"Salah: {gm.WrongAnswers}";
        }

        // Pesan motivasi
        if (motivationText)
        {
            if (!levelPassed)
                motivationText.text = "Jangan menyerah! Coba lagi ya, kamu pasti bisa! 🤞";
            else if (gm != null && gm.WrongAnswers == 0)
                motivationText.text = "Luar biasa! Kamu menguasai materi ini dengan sempurna! ⭐";
            else
                motivationText.text = "Kerja bagus! Terus belajar dan tingkatkan pemahamanmu! 👍";
        }

        // Atur tombol-tombol
        SetupResultButton(nextLevelButton, levelPassed, onNextLevel);
        SetupResultButton(retryButton, true, onRetry);
        SetupResultButton(mainMenuButton, true, onMainMenu);
    }

    private void SetupResultButton(Button btn, bool active, Action callback)
    {
        if (btn == null) return;
        btn.gameObject.SetActive(active);
        btn.onClick.RemoveAllListeners();
        btn.onClick.AddListener(() => callback?.Invoke());
    }
}

```

Assets/Scripts/UI/PauseMenu.cs

Menu jeda (Esc).

```

using UnityEngine;
using UnityEngine.InputSystem; // Input System baru
using EduQuest.Core;
using EduQuest.Gameplay;

namespace EduQuest.UI
{

```



```

/// <summary>
/// Pause Menu sesuai GDD: Resume, Restart, Main Menu.
/// Tombol Esc untuk toggle. Tidak bisa pause saat quiz sedang aktif.
/// </summary>
public class PauseMenu : MonoBehaviour
{
    [Header("Panel Pause")]
    public GameObject pausePanel;

    public bool IsPaused { get; private set; }

    private void Start()
    {
        if (pausePanel) pausePanel.SetActive(false);
    }

    private void Update()
    {
        // Deteksi tombol Esc (Input System baru)
        bool escPressed = Keyboard.current != null && Keyboard.current.escapeKey.wasPressedThisFrame;

        // Jangan boleh pause saat quiz berlangsung
        bool quizActive = QuizManager.Instance != null && QuizManager.Instance.IsQuizActive;

        if (escPressed && !quizActive)
        {
            if (IsPaused) Resume();
            else Pause();
        }
    }

    public void Pause()
    {
        IsPaused = true;
        if (pausePanel) pausePanel.SetActive(true);
        Time.timeScale = 0f; // hentikan game
        Cursor.lockState = CursorLockMode.None;
        Cursor.visible = true;
    }

    public void Resume()
    {
        IsPaused = false;
        if (pausePanel) pausePanel.SetActive(false);
        Time.timeScale = 1f;
        Cursor.lockState = CursorLockMode.Locked;
        Cursor.visible = false;
    }

    public void Restart()
    {
        Time.timeScale = 1f;
        SceneLoader.Instance?.ReloadCurrentScene();
    }

    public void GoToMainMenu()
    {
        Time.timeScale = 1f;
        SceneLoader.Instance?.LoadMainMenu();
    }
}

```

}

}

9. Modul EndScreen

A. Langkah Membangun

New Scene > **EndScreen**. Main Camera Background #0A0F0A. GameManagers (GameManager + SceneLoader). Canvas + Panel (#0F170F).

Elemen	Anchor	Pos	Size	Font/Warna
Title "SELAMAT!"	TC	0, -120	1200×130	90 putih
Skor Akhir	C	0, 160	900×80	55 #FFE666
Statistik	C	0, 80	900×60	40 putih
Pesan	C	0, -40	1000×160	32 #D9F2D9
Button Main Lagi (→OnPlayAgain)	C	0, -260	450×80	—
Button Menu Utama (→OnMainMenu)	C	0, -360	450×80	—

Create Empty EndScreenController > Add EndScreenController; assign titleText/scoreText/statsText/messageText.

B. Kode Modul EndScreen

Assets/Scripts/UI/EndScreenController.cs

Layar akhir.

```
using UnityEngine;
using TMPro;
using EduQuest.Core;

namespace EduQuest.UI
{
    /// <summary>
    /// Layar akhir (EndScreen) setelah semua level selesai / game over.
    /// Menampilkan skor total dan kondisi menang/kalah sesuai GDD.
    /// </summary>
    public class EndScreenController : MonoBehaviour
    {
        [Header("Teks")]
        public TMP_Text titleText;
        public TMP_Text scoreText;
        public TMP_Text statsText;
        public TMP_Text messageText;

        private void Start()
        {
            Cursor.lockState = CursorLockMode.None;
            Cursor.visible = true;
            Time.timeScale = 1f;

            var gm = GameManager.Instance;
            if (gm == null) return;

            bool win = !gm.IsGameOver; // kalah jika nyawa habis

            if (titleText) titleText.text = win ? "SELAMAT! 🏆" : "GAME OVER";
            if (scoreText) scoreText.text = $"Skor Akhir: {gm.Score}";
            if (statsText) statsText.text = $"Benar: {gm.CorrectAnswers} | Salah: {gm.WrongAnswers}";

            if (messageText)
            {
            }
```

```
        messageText.text = win
        ? "Kamu berhasil menjelajahi seluruh sistem organ manusia!\nTeruslah belajar dan jaga kesehatanmu! 🍌"
        : "Nyawamu habis. Jangan menyerah, coba lagi ya! 💙";
    }
}

/// <summary>Tombol Main Lagi: reset & kembali ke menu.</summary>
public void OnPlayAgain()
{
    GameManager.Instance?.ResetGame();
    SceneLoader.Instance?.LoadMainMenu();
}

public void OnMainMenu()
{
    SceneLoader.Instance?.LoadMainMenu();
}

public void OnQuit()
{
    Application.Quit();
}
}
```

10. Build Settings & WebGL

File > Build Settings > Add Open Scenes berurutan:

MainMenu → Level1_Jantung → Level2_ParuParu → Level3_Otak → EndScreen

WebGL: Switch Platform; Color Space **Gamma**; Compression **Gzip**. Application.Quit() tidak berefek di WebGL.

11. Troubleshooting

Masalah	Solusi
Tombol UI tak bisa diklik	EventSystem → InputSystemUIInputModule.
Teks UI tak tampil	Import TMP Essentials.
Kamera tak berputar	Set Follow PlayerFollowCamera ke PlayerCameraRoot.
Duplikat ThirdPersonController	Hapus salah satu.
.glb tak bisa di-drag	Install glTFast atau konversi ke FBX.
Stasiun tak munculkan prompt E	Box Collider Is Trigger; player Tag "Player".

— Selesai. Tutorial lengkap EduQuest. Selamat berkarya! —